

Supplementary Materials for
**Stimulated Raman Photothermal Microscopy Towards Ultrasensitive
Chemical Imaging**

Yifan Zhu *et al.*

Corresponding author: Ji-Xin Cheng, jxcheng@bu.edu

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Movies S1 and S2

Supplementary Text

The SRP Ball lens model

The model is defined in pure liquid under the SRS excitation of pump and Stokes laser pulses with photon flux ϕ (photon $\cdot$ cm $^{-2}\cdot$ s $^{-1}$) defined as:

$$\phi = \frac{P}{h\nu \cdot A} \cdot \frac{1}{f_{rep}\tau_{exc}} \quad (S1)$$

where the P is average laser power, h is the Planck constant, ν is the laser frequency, f_{rep} is the laser repetition rate, and τ_{exc} is the laser pulse width. The number of SRS events (N_{SRS}) induced by a single pair of pump and Stokes pulses is given by:

$$N_{SRS} = N_{mol} \cdot \sigma_{SRS} \cdot \phi_{pump} \cdot \phi_{Stokes} \cdot \tau_{exc} \quad (S2)$$

where N_{mol} is the number of Raman active chemical bonds in the laser focus, σ_{SRS} is the stimulated Raman cross-section (unit in cm $^4\cdot$ s $\cdot$ photon $^{-1}$), and ϕ_{pump} and ϕ_{Stokes} are the photon flux of pump and Stokes lasers, respectively.

The energy deposition (Q) through the SRS process is then given by:

$$Q = \hbar\omega_{SRS} \cdot N_{SRS} \quad (S3)$$

where $\hbar$ is the reduced Planck constant and ω_{SRS} is the Raman transition energy.

Under the approximation that SRS excitation creates uniform heating in a spherically symmetric region, with its radius (r) defined as half of the lateral resolution of SRS excitation:

$$r = \frac{1}{4NA} \sqrt{1 / \left(\frac{1}{\lambda_{pump}^2} + \frac{1}{\lambda_{Stokes}^2} \right)} \quad (S4)$$

and excitation volume:

$$V_{SRS} = \frac{4}{3}\pi r^3 \quad (S5)$$

In this region, energy deposition causes a temperature rise (ΔT):

$$\Delta T = \frac{Q}{C_p \cdot V_{SRS}} \quad (S6)$$

where C_p is the heat capacity at constant pressure of the medium and V_{SRS} is the volume of the SRS excitation. Thermal diffusion is not considered as the energy deposition of a single pair of laser pulses is finished within the thermal confinement time (~ 1 μ s). The pressure is assumed to be constant as the stress confinement time (< 1 ns) is too short to be captured by the current measurement system (20 MHz).

The elevated temperature then changes the refractive index of the medium (Δn):

$$\Delta n = \Delta T \cdot \frac{dn}{dT} \quad (S7)$$

where dn/dT is the thermo-optic coefficient of the medium.

As shown in **Fig. S4**, the changed refractive index effectively forms a thermal ball lens, with its focal length f satisfying the relationship that:

$$\frac{1}{f} = \frac{2}{r} \left(\frac{n_1}{n_0} - 1 \right) = \frac{2\Delta n}{n_0 r} \quad (S8)$$

Here, r is the radius of the ball lens, n_0 is the refractive index of the medium, and n_1 is the refractive index of the ball lens.

To probe the thermal lens effect, we introduce a focused Gaussian probe laser that is collinear with the laser beams for SRS. The ball lens changes the propagation of the probe light, with the dashed lines indicating the light propagation without a ball lens, and the solid lines indicating light propagation with a ball lens:

$$\frac{1}{f} = \frac{1}{q} - \frac{1}{z_f} \quad (S9)$$

where z_f denotes the offset between the SRS and probe laser focus.

Under the paraxial approximation, for an annular ray of light with incident angle θ as shown in **Fig. S3**, Eq. (S9) becomes:

$$\frac{1}{f} \approx \frac{\Delta\theta(\theta)}{z_f \sin\theta \cos\theta} \quad (S10)$$

Here, $\Delta\theta(\theta)$ is the change of propagation angle of the probe light at angle θ .

Combining Eq. (S8) and Eq. (S10) yields the change of laser propagation angle $\Delta\theta$ induced by the ball lens as a function of incident angle θ :

$$\Delta\theta(\theta) = \frac{2z_f \cdot \Delta n \cdot \sin\theta \cos\theta}{n_0 r} \quad (S11)$$

The far-field annular intensity distribution of the probe laser, $I(\theta)$, can be written as

$$I(\theta) = I_0 \pi Z_R^2 \cdot \theta \cdot \exp\left(-\frac{Z_R^2}{\omega_0^2} \cdot \theta^2\right) \quad (S12)$$

where Z_R is the Rayleigh length of the laser focus and ω_0 is the beam waist. After propagating through the ball lens, the probe laser intensity is measured by a photodetector with angular detection aperture size θ_{Det} . Thus, the SRP signal is calculated as the intensity difference with the presence or absence of the ball lens:

$$I_{SRP} = \int_0^{\theta_{Det}} I(\theta + \Delta\theta(\theta)) d\theta - \int_0^{\theta_{Det}} I(\theta) d\theta \approx I(\theta_{Det}) \cdot \Delta\theta(\theta_{Det}) \quad (S13)$$

Combining all above equations gives:

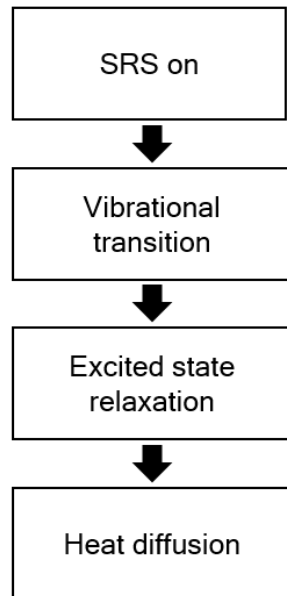
$$I_{SRP} \approx I(\theta_{Det}) \frac{4z_f \cdot \sin\theta_{Det} \cdot \cos\theta_{Det}}{n_0 r} \cdot \frac{1}{C_p} \frac{dn}{dT} \cdot \frac{1}{V_{SRS}} \cdot (\hbar\omega_{SRS}) (N_{mol}\sigma_{SRS}\phi_{pump}\phi_{Stokes}\tau_{exc}) \quad (S14)$$

In Eq. (S14), $\frac{1}{C_p} \frac{dn}{dT}$ denotes the thermal and thermo-optic properties of the sample environment. V_{SRS} denotes the volume of heating. $\hbar\omega_{SRS}$ represents the energy of vibrational transition. $N_{mol}\sigma_{SRS}\phi_{pump}\phi_{Stokes}\tau_{exc}$ represents the number of transitions caused by SRS

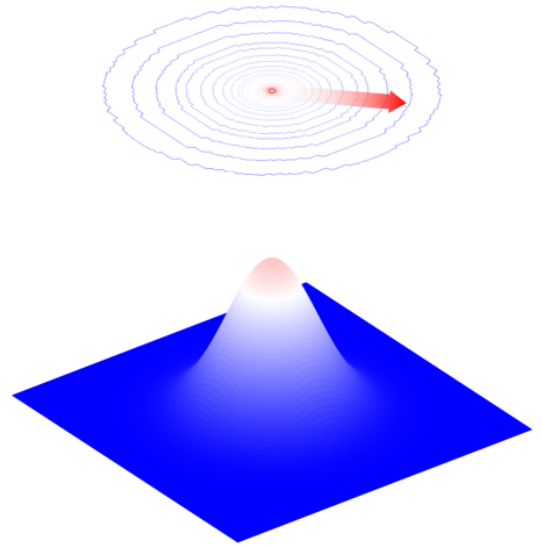
Under the tight focusing condition, a FDTD simulation is conducted to calculate the dependence of SRP intensity on the molecular concentration. First, a thermal ball lens model is generated in

MATLAB as a matrix, where the refractive index outside the ball lens is set to the refractive index of the DMSO medium ($n_0 = 1.4793$), and the refractive index inside the ball lens region is set to different lower values ($n = n_0 + \Delta n$). Then, the thermal ball lens model is imported into FDTD as an “(n,k) material” object. A Gaussian focused light source is sent through the thermal ball lens object and is measured with a far-field intensity analyzer, of which the output is a far-field angular distribution of intensity. The middle 31° (~ 0.4 NA) of the angular distribution of intensity is integrated, then subtracted by the corresponding 0-concentration value to determine the SRP signal intensity.

A



B



$$dQ = -kA \frac{dT}{dr} = -k(4\pi r^2) \frac{dT}{dr}$$

Fig. S1. Simulation modal for SRP induced thermal effect. (A) Thermal effect generation scheme. (B) Schematic of a heat diffusion model. The system is meshed into concentric spherical shells. The red arrow indicates the thermal propagation direction.

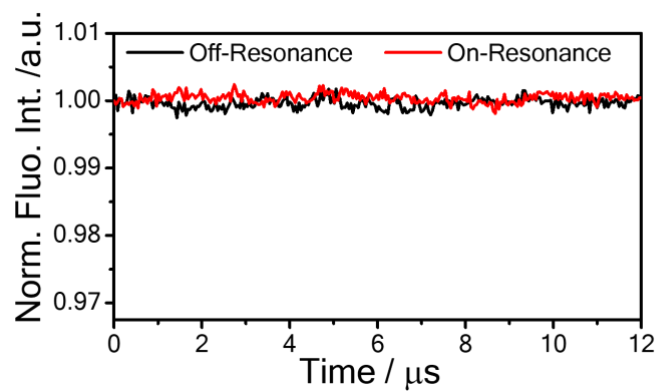


Fig. S2. Measured fluorescence of rhodamine 800 in DMSO during an SRS process. The beating frequency ($\omega_p - \omega_s$) was tuned to 2913 cm^{-1} for *On-Resonance* and 2850 cm^{-1} for *Off-Resonance*.

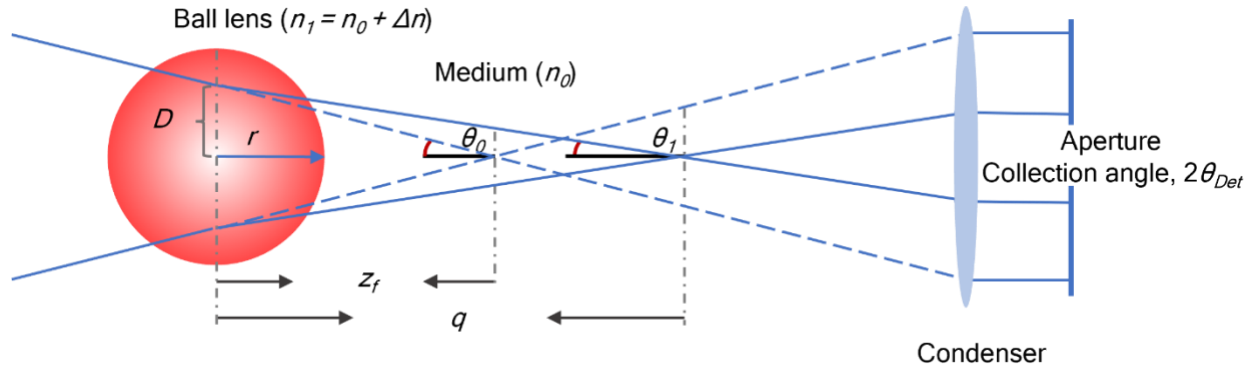


Fig. S3. Thermal ball lens model. The solid line indicates the propagation in the absence of a ball lens. The dashed line indicates the propagation of the probe laser in the presence of a ball lens. n_0 is the refractive index of the medium. Δn is the change of refractive index induced by the SRP effect.

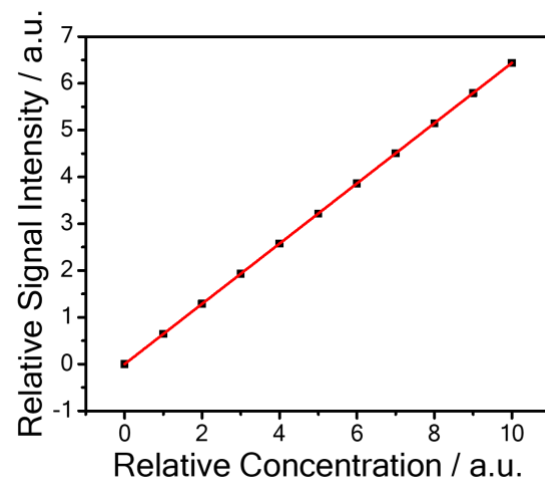


Fig. S4. Finite-difference time-domain simulation of SRP in high NA condition shows linear relationship between relative concentration and SRP signal intensity. Probe laser illumination NA: 1.47; collection NA: 0.4; simulation wavelength: 520 nm.

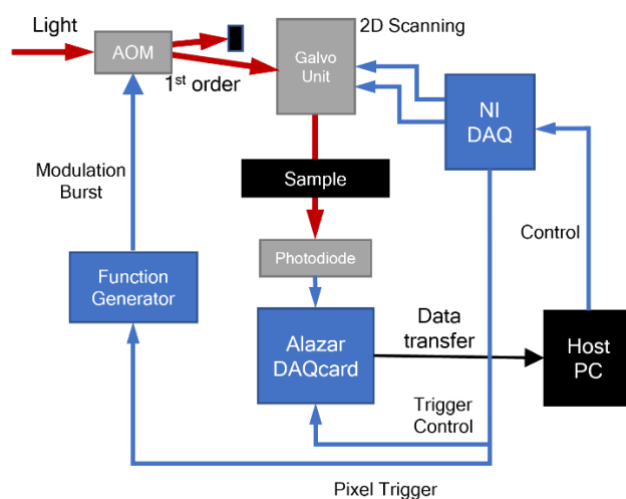


Fig. S5. Electronic control and data acquisition flow in the SRP microscope.

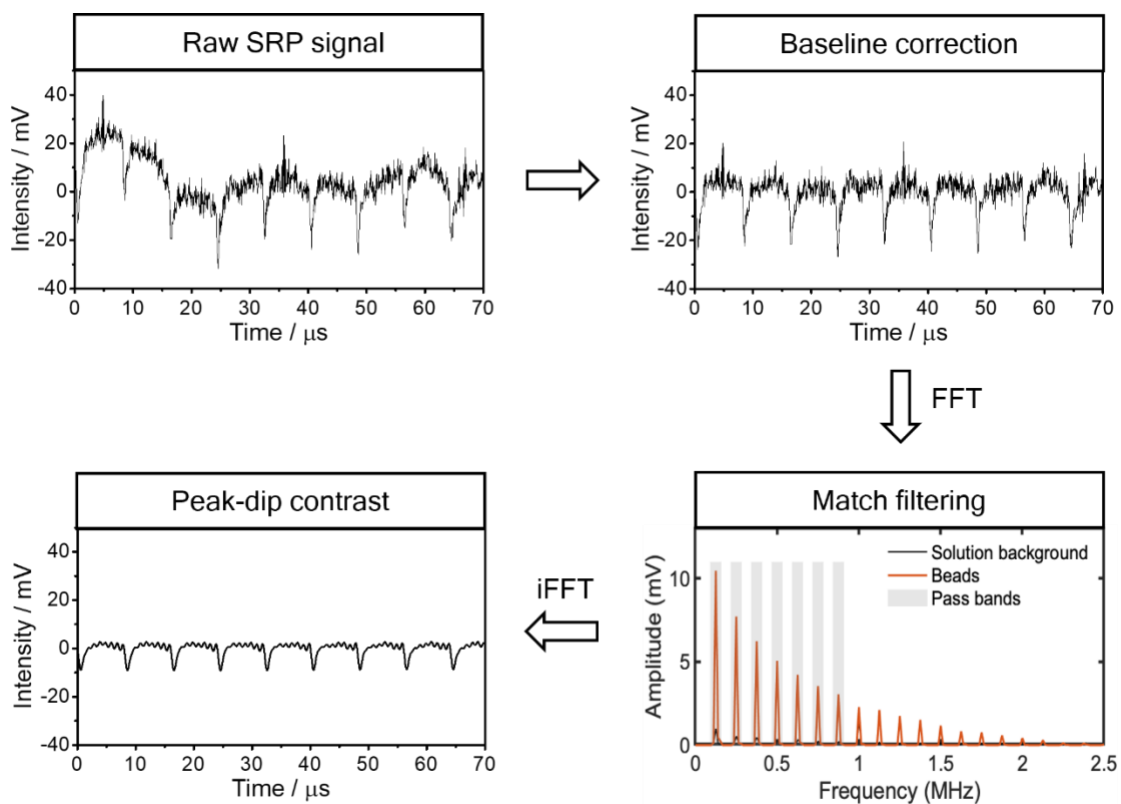


Fig. S6. Data processing of a thermal dynamic trace recorded by an SRP microscope. The signal is acquired from a single 100-nm PMMA nanoparticle at Raman shift of 2950 cm^{-1} immersed in glycerol- d_8 . FFT: fast Fourier transform; iFFT: inverse fast Fourier transform.

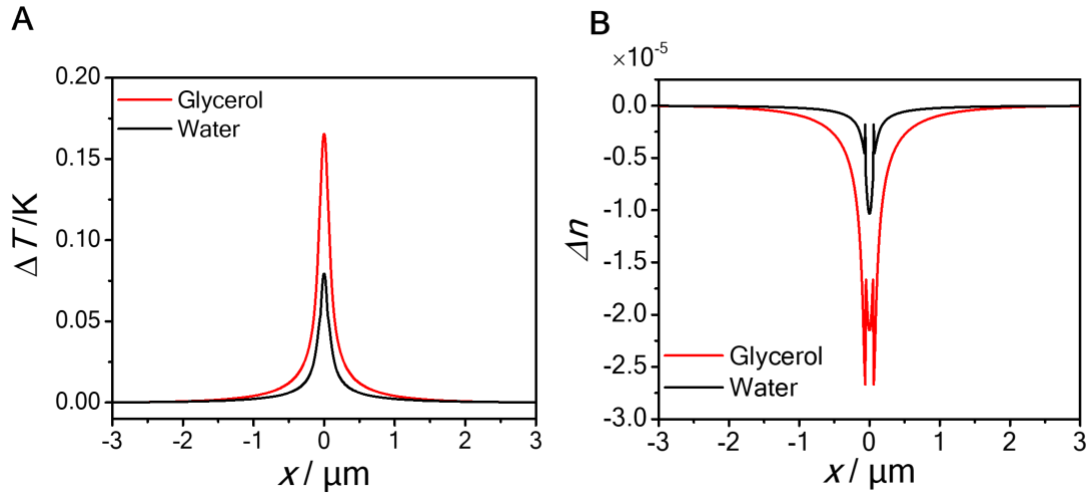


Fig. S7. Simulation results for the shape of temperature gradient (A) and thermal lens (B) induced by the SRS absorption of a 100-nm PMMA particle in glycerol or water. The PMMA particle is placed at $x = 0$. Two local minimums of refractive index change appear at two PMMA-medium boundaries, due to the sudden change of thermos-optic coefficients (dn/dT) at the boundaries.

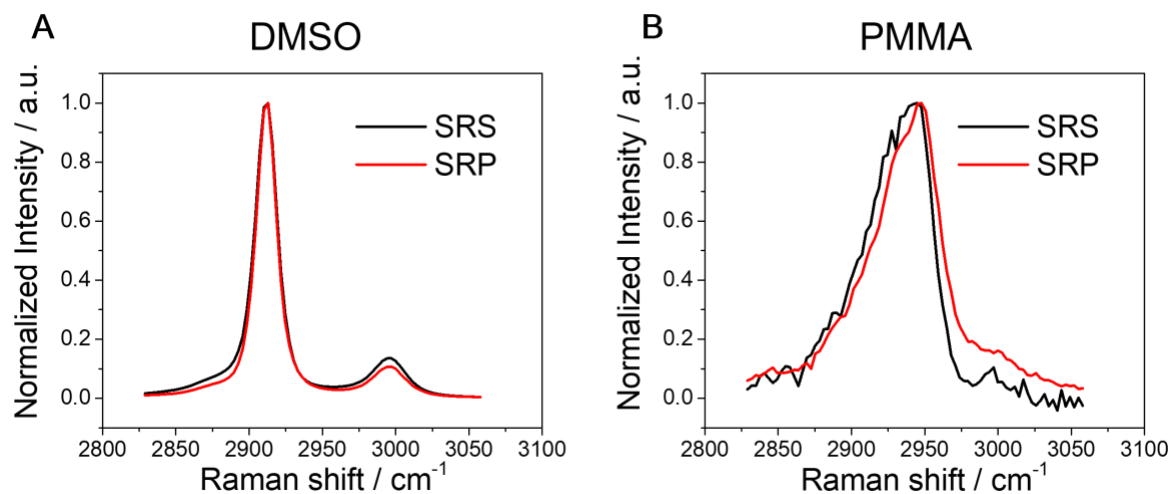


Fig. S8. SRP spectral fidelity. (A) Comparison between SRS and SRP spectra of pure DMSO. (B) Comparison between SRS and SRP spectra of a 500-nm PMMA particle.

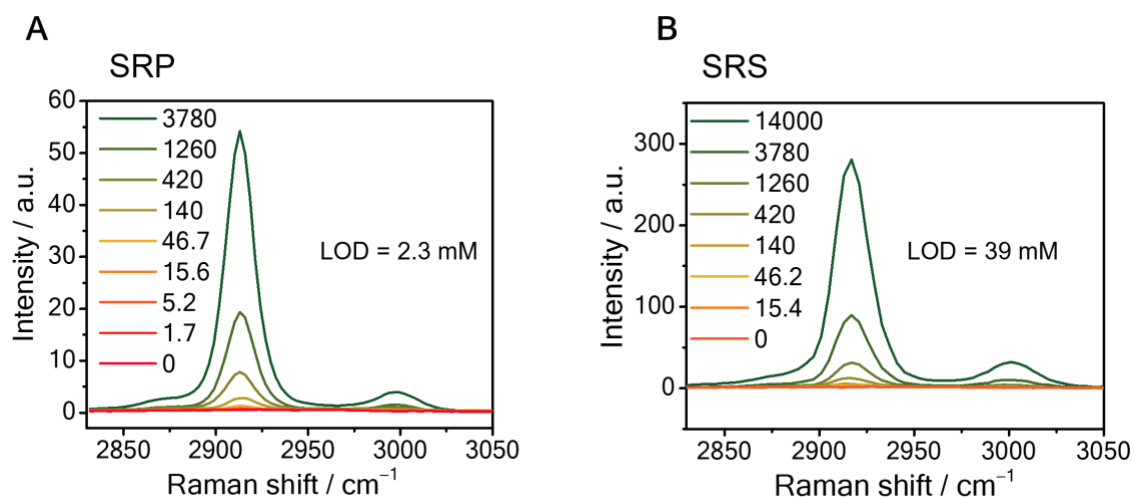


Fig. S9. Complete data set of LOD measurement of DMSO dissolved in DMSO-d6 with SRP (A) and SRS (B). Unit of concentration: mM.

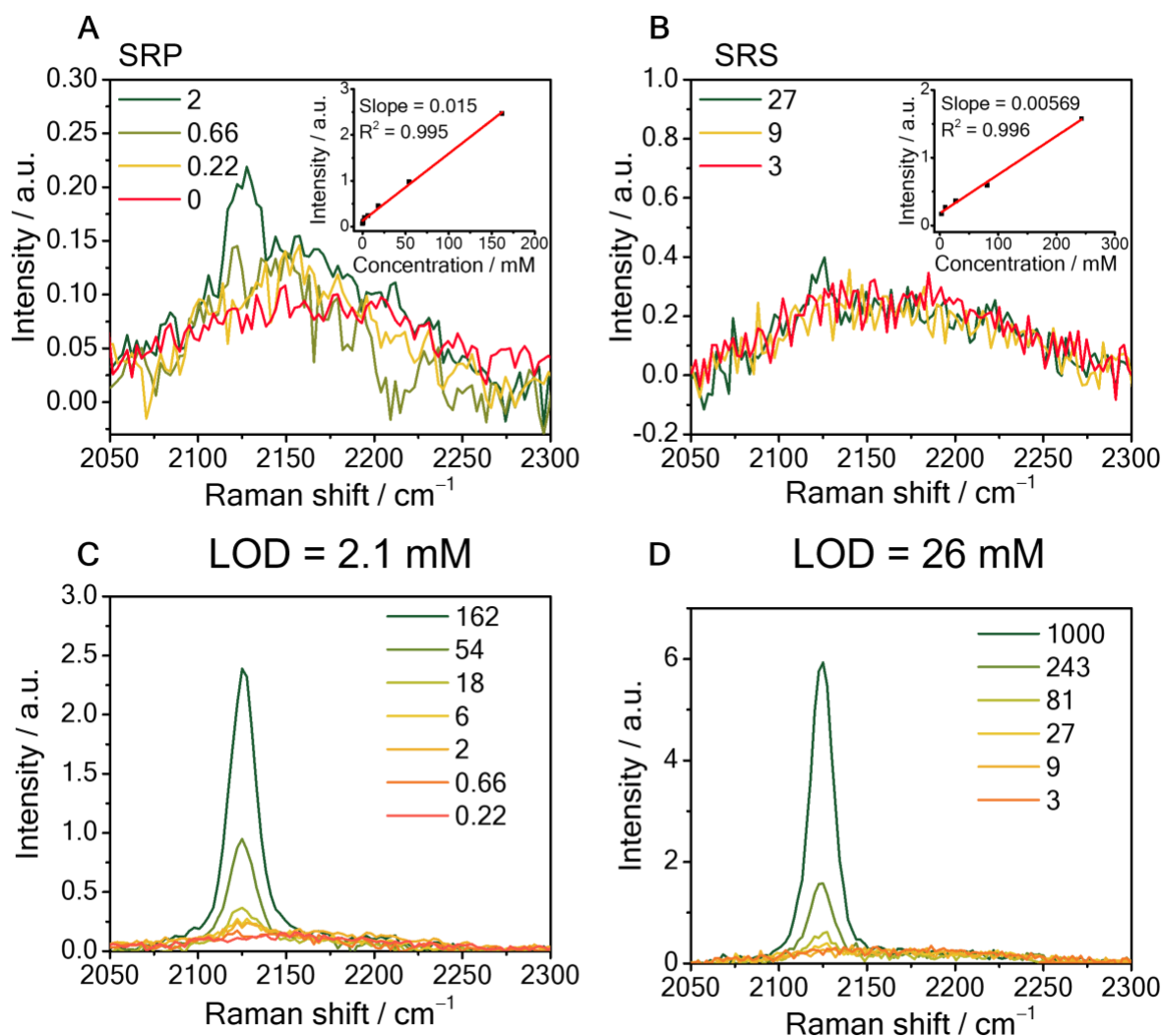


Fig. S10. Limit of detection (LOD) of SRP and SRS for 1,7-Octadiyne in DMSO. (A-B) SRP (A) or SRS (B) signal with a gradient concentration of 1,7-Octadiyne dissolved in DMSO. Insert shows the signal intensity as a function of concentration. (C-D) The complete data set of LOD measurement with SRP (C) and SRS (D). Unit of concentration: mM.

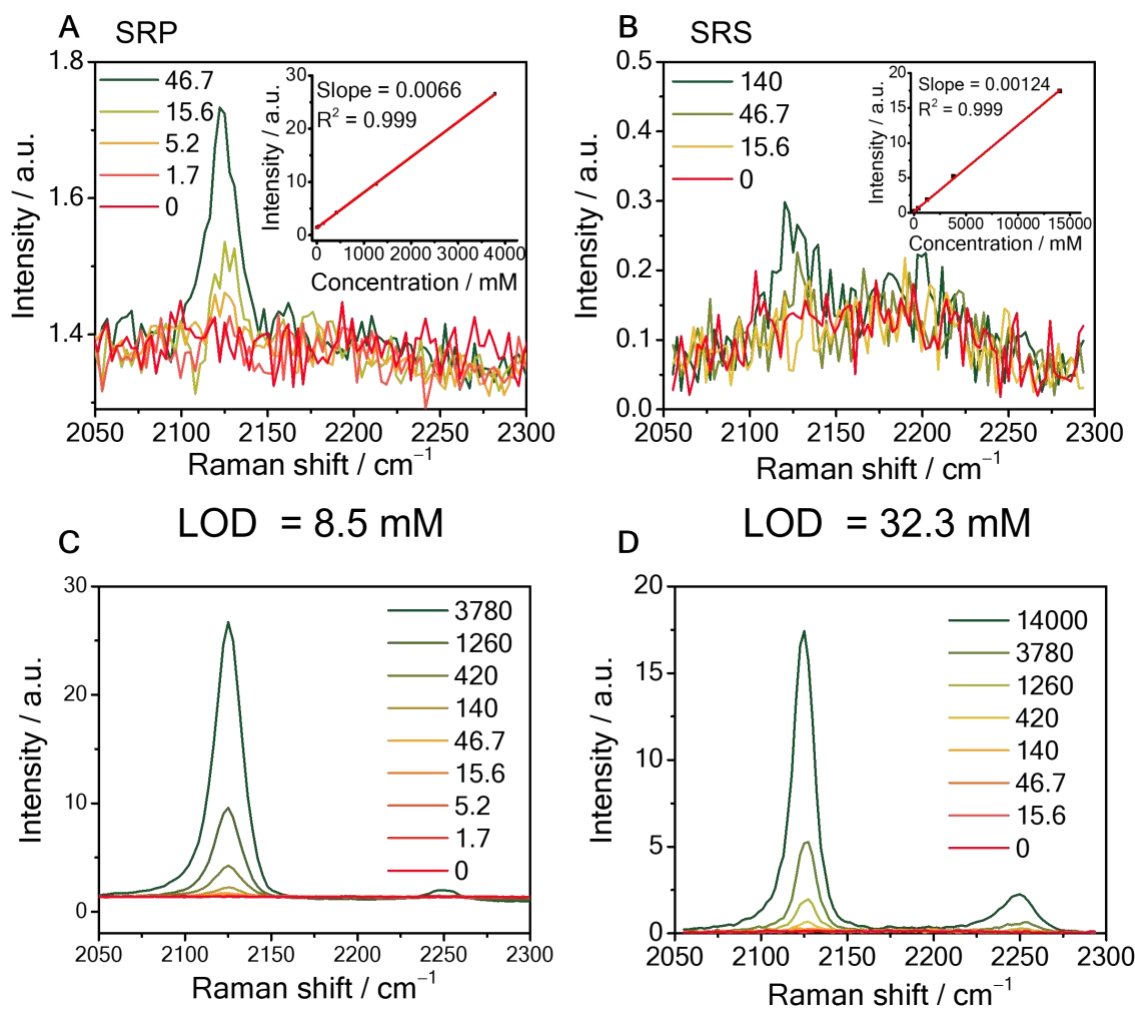


Fig. S11. Limit of Detection (LOD) of SRP and SRS for DMSO-d₆ in DMSO. (A-B) SRP (A) or SRS (B) signal with a gradient concentration of DMSO-d₆ dissolved in DMSO. Concentration unit: mM. Insert shows the signal intensity as a function of concentration. (C-D) The complete data set of LOD measurement with SRP (C) and SRS (D).

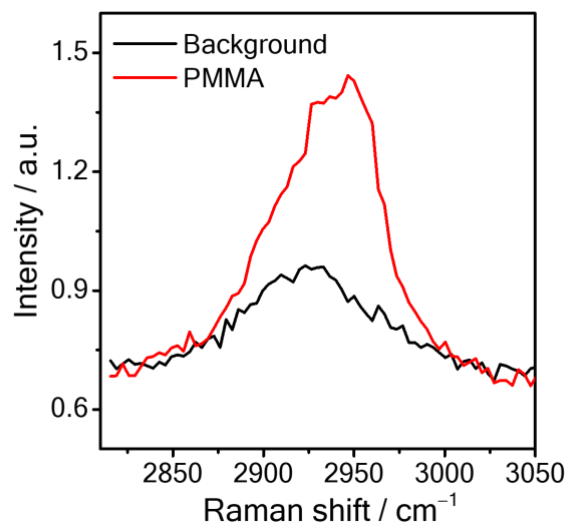


Fig. S12. SRP spectrum of a single 100 nm PMMA nanoparticle immersed in glycerol. The peak around 2950 cm⁻¹ is from CH₃ stretching vibration in PMMA.

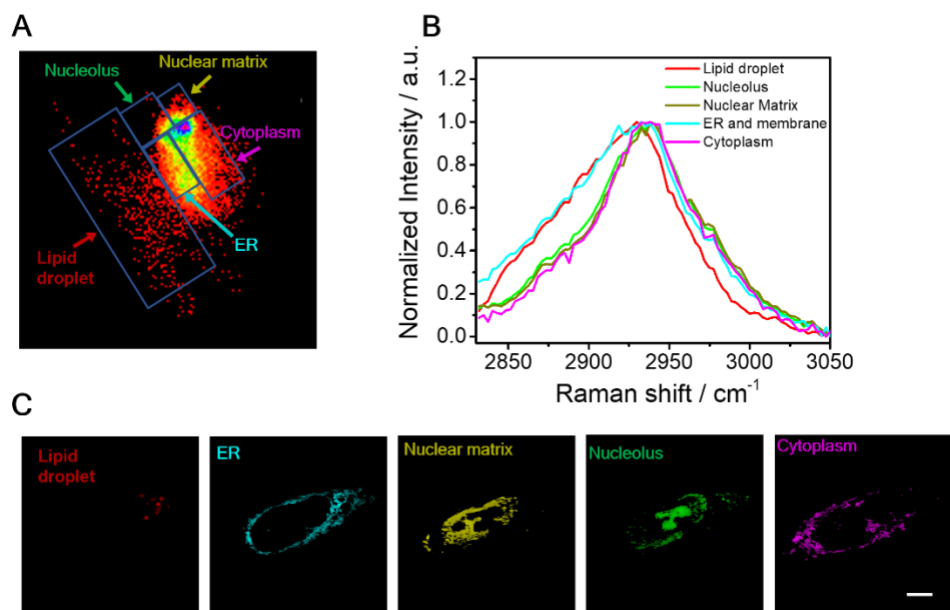


Fig. S13. Phasor analysis for the SRP image of SJS-A-1 cells in aqueous environment. (A) Segmentation of the SRP image in the phasor domain. (B) The SRP spectrum at C-H region for each component. (C) Mapping of each component from phasor analysis. Scale bar: 10 μm .

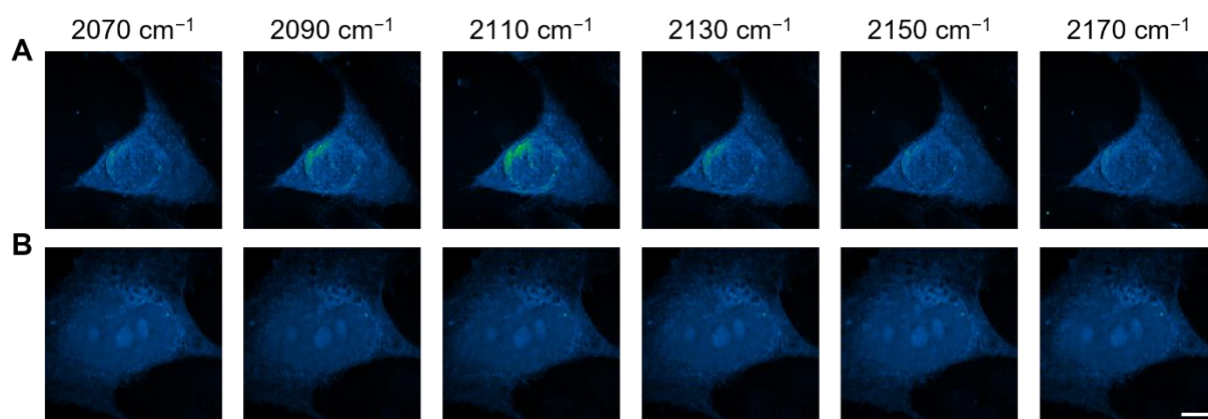


Fig. S14. Hyperspectral SRP imaging of SJSA-1 cell in PBS. Data were shown with 20 cm^{-1} spectral step. The cells were incubated with (A) or without (B) deuterated palmitic acid (PA-d31) for 18 h prior to imaging. Scale bar: $10 \mu\text{m}$.

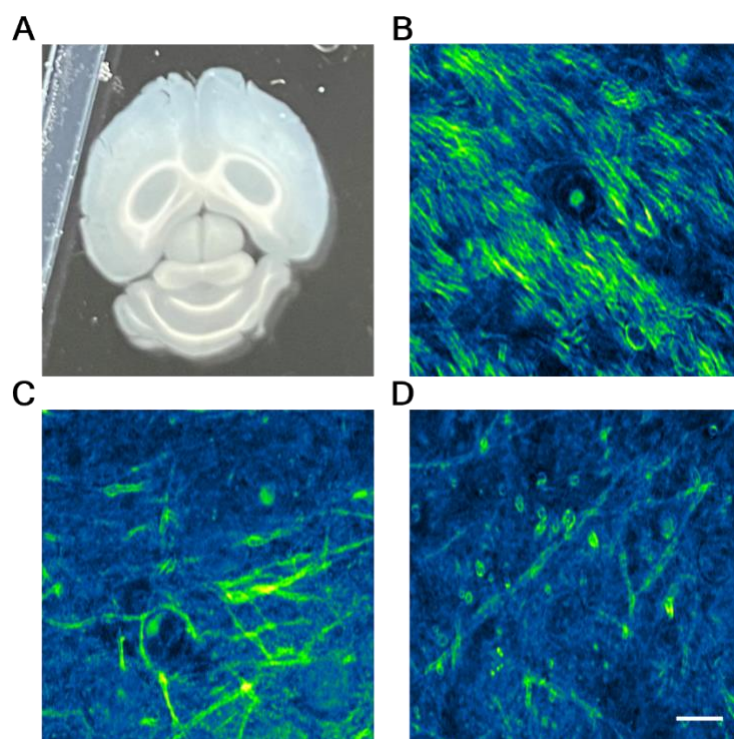


Fig. S15. SRP imaging of a mouse brain slice immersed in PBS. (A) Picture of the brain slice sample. (B-D) Representative SRP images of the brain slice sample at 2930 cm^{-1} . Scale bar = 12 μm .

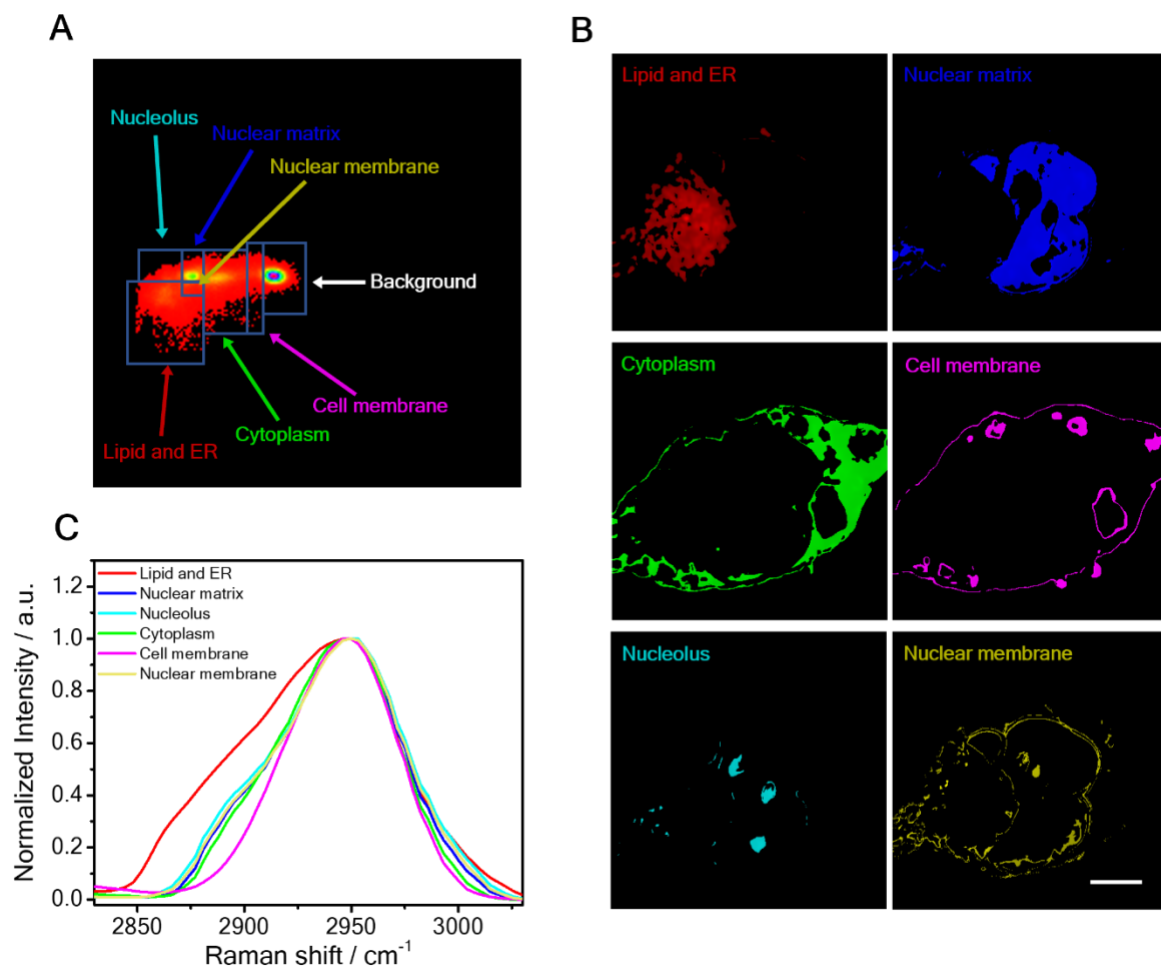


Fig. S16. Phasor analysis for the SRP image of glycerol-d8 immersed Mia PACA-2 cells. (A) Segmentation of the SRP image in the phasor domain. (B) Maps of each component from phasor analysis. Scale bar = 5 μm . (C) The SRP spectrum at C-H region for each component.

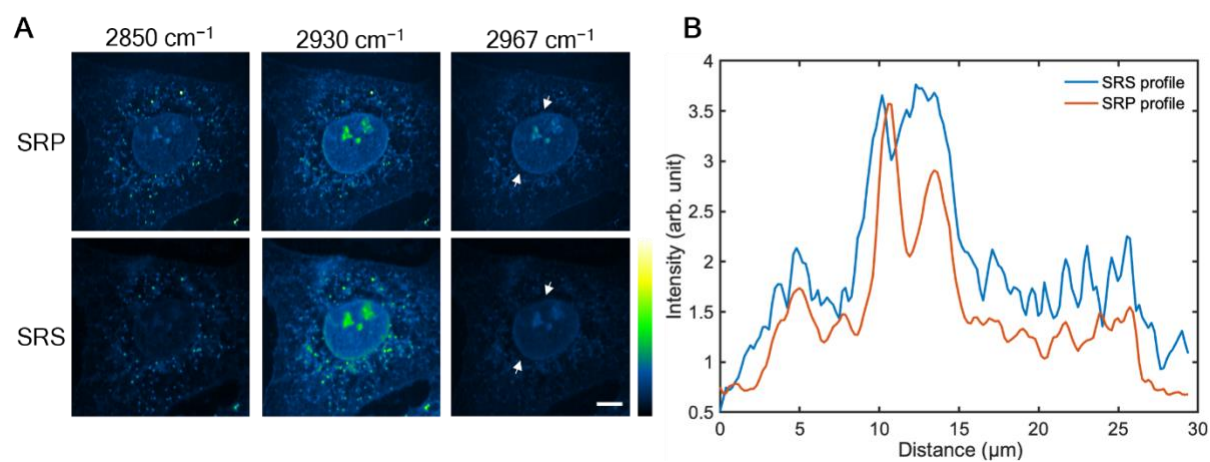


Fig. S17. Comparison between SRP and SRS images of SKOV3 cell immersed in glycerol-d8. (A) SRP and SRS images taken at three Raman shifts for lipid, protein and DNA. Scale bar: 10 μm . (B) The intensity profiles along the arrow-indicated lines in the SRP and SRS images at 2967 cm^{-1} .

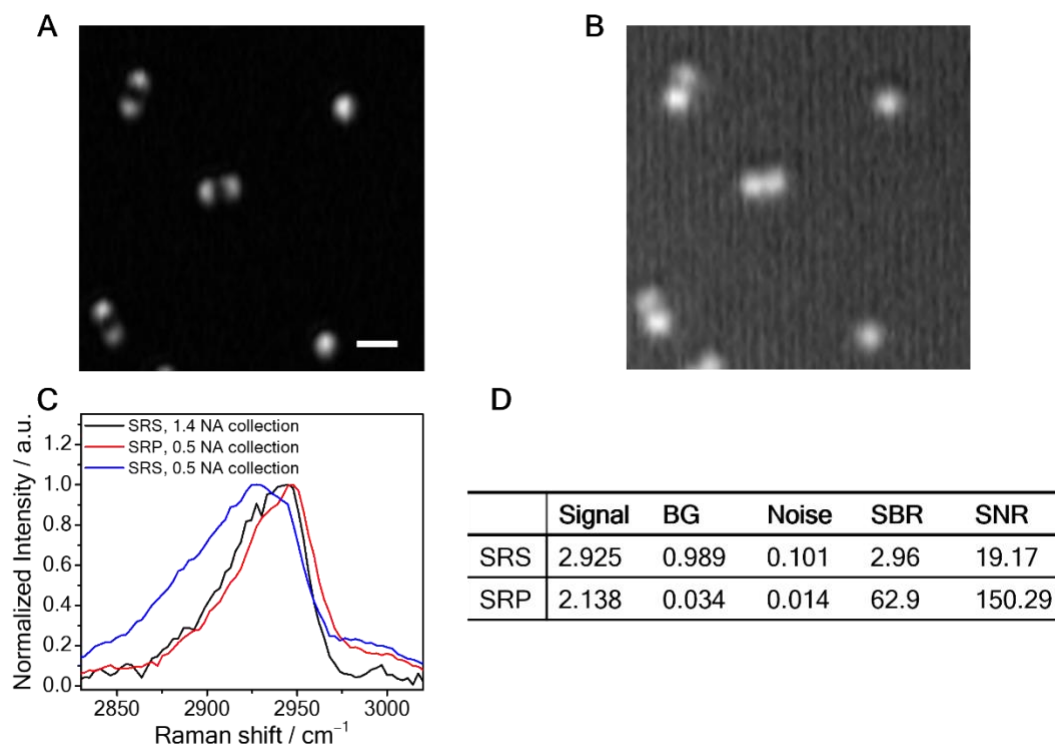


Fig. S18. Compare SRP and SRS performance under different light collection condition. SRP (A) and SRS (B) images of 500 nm PMMA nanoparticles recorded with 1.2 (NA) water objective for focusing and a 0.5 NA air condenser for signal collection. Raman shift: 2950 cm⁻¹. Scale bar: 1.5 μ m. (C) Comparison between the SRP and SRS spectra of a single 500 nm PMMA nanoparticle. The broader line profile in SRS with low NA collection (blue line) is caused by a Raman-independent cross-phase modulation background. (D) Image quality comparison between SRP and SRS images. BG: background. SBR: signal-to-background ratio.

Thermal property	Unit	DMSO	hexane	glycerol	PMMA	water
Heat Capacity	J/(kg·K)	1966	2251	2400	1466	4184
Thermal conductivity	W/(m·K)	0.200	0.124	0.283	0.167-0.25	0.598
Thermo-optic coefficient dn/dT (10 ⁻⁴)	K ⁻¹	-4.93 (34)	-5.20 (51)	-2.30 (52)	-1.30 (53)	-1.13 (40)
Refractive index	-	1.479	1.375	1.473	1.492	1.333
Relative signal intensity	a.u.	8.37	8.29	3.21	2.93	1

Table S1. Thermal and thermo-optic properties and estimated relative signal intensity of the materials related to the study.

Movie S1. SRP imaging of lipid dynamics inside a living SJSA-1 cell. Frame rate: 2.8 s/frame. Scale bar: 5 μm .

Movie S2. Hyperspectral C-H SRP imaging of fixed SJSA-1 cell in aqueous environment. Scale bar: 10 μm .

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